Analytical exam

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1) $P = HELLO$

$K = 15$

Encryption:-

Ceaser cipher $(C) = (P+K) \% 26$

i) $P = H = 7$

$C = (7+15) \% 26 = 22 = W$

ii) $E = 4$

$C = (4+15) \% 26 = 19 = T$

iii) $L = 11$

$C = (11+15) \% 26 = 0 = A$

iv) $L = 11$

$C = (11+15) \% 26 = 0 = A$

v) $O = 14$

$C = (14+15) \% 26 = 3 = D$

$C = WTAAD$

Decryption:

$P = (C - k) \% 26$

i) $C = W = 22$

$P = (22 - 15) \% 26 = 7 = H$

ii) $C = T = 19$

$P = (19 - 15) \% 26 = 4 = E$

iii) $C = A = 0$

$P = (0 - 15) \% 26 = L$

iv) $C = A = 0$

$P = (0 - 15) \% 26 = L$

v) $C = D = 3$

$P = (3 - 15) \% 26 = O$

2) P = meet me after the Party

k = 3

Encryption:-

~~m = p, e = h, t, w~~

M = p

~~m~~e = h

e = h

t = w

m = e

e = h

a = d

f = i

t = w

e = h

r = v

t = w

h = k

e = h

p = s

a = d

r = v

t = w

y = b

C = Phhw ehdiwhvw khsdvwb

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P = Cryptography

Mono alphabetic:-

A	B	C	D	E	F	G	H	I	J
Z	Y	X	W	V	U	T	S	R	Q
K	L	M	N	O	P	Q	R	S	T
P	O	N	M	L	K	J	I	H	G
U	V	W	X	Y	Z				
F	E	D	C	B	A				

Encryption:-

C = XIBKGLTIZSB

Meet Me TOMORROW

k = keyWORD

Play fair:-

K	E	Y	W	O
R	D	A	B	C
F	G	H	I/J	L
M	N	P	Q	S
T	U	V	X	Z

Me = NK

et = KU

me = NK

TO = ZK

mo = SK

rx = BT

ro = CK

wx = BW

C = NKKUNKZKSKBTCKBW

5) P = ACTIVE

K = PASCAL

Vignere cipher :- $C = (P_i + K_i) \% 26$

Encryption :-

C ⇒ A + P = P

C + A = C

T + S = L

I + C = K

V + A = V

E + L = P

C = PCLKVP

Decryption :-

P ⇒ P − P = A

C − A = C

L − S = T

K − C = I

V − A = V

P − L = E

P = ACTIVE

6) P = HELP

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

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Hill cipher :-

$$C = \begin{bmatrix} H & E \\ L & P \end{bmatrix} * \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \% 26 = \begin{bmatrix} 7 & 4 \\ 11 & 15 \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \% 26$$

$$= \begin{bmatrix} 3 & 15 \\ 11 & 4 \end{bmatrix} = \begin{bmatrix} D & P \\ L & E \end{bmatrix}$$

C = DPLE